ENTRY 041 The First GNN — August 16, 2026 — Status: COMPLETE

Entry 041 — A Graph Neural Network Cuts the Floor-Collapse Error by 63%

Real Route Structure, Not Hand-Picked Summaries, Is What the Tabular Model Was Missing

1. Architecture

Each circuit's real physical route (Entry 031/032's route-then-entangle structure) is represented as a small graph: nodes are the physical qubits the transpiler actually touched (2-37 per circuit), node features are that qubit's own T1/T2/readout; edges are the real 2-qubit gates issued (SWAPs and the final entangling gate), edge features are that edge's gate error and duration. Two rounds of message passing (16-dim hidden state, mean aggregation) let information several hops away reach a node's embedding, then a masked mean-pool over nodes feeds a small MLP alongside chip/kind/hop-distance and v4.1's own prediction (included as an input feature, same as Entry 037's tabular baseline, so the comparison is fair). Built in JAX + Optax (no PyTorch available in this environment); graphs are padded to 40 nodes/edges with masks so the whole batch vmaps. entry041_build_graphs.py re-derives the exact graph for all 1,376 Entry 034-040 circuits from their stored (chip, qubit pairs) via route_with_explicit_swaps -- no new simulation needed, the labels already exist.

2. Result: A Real, Large Improvement

model	CV MAE (all 1,376)	CV R^2	MAE, floor-collapse only
v4.1 (closed form)	4.50 pts	n/a	17.01 pts
Gradient boosting (Entry 037/040)	3.92 pts	0.785	13.78 pts
GNN (this entry)	2.64 pts	0.868	6.22 pts

The GNN beats the tabular model everywhere: 33% lower overall MAE (3.92 to 2.64), and on the 70 floor-collapse cases specifically, a 63% reduction from v4.1's 17.01 points down to 6.22 -- more than triple the tabular model's ~20% relative gain. Spot-checking individual floor-collapse predictions confirms this isn't an aggregate artifact: on 13 of 15 sampled cases the GNN lands within 5 points of the true ~50% floor, while v4.1 predicts 80-91% on several of the same circuits.

3. Why the Graph Structure Helps

Entry 036 showed floor-collapse comes from correlated noise across a route that no per-channel closed form captures, and that route length alone doesn't predict it -- a 3-edge route produced the single largest independence-breakdown gap measured in the project. The tabular model's features (worst edge error, worst T1, hop count) are scalar summaries that throw away exactly the arrangement information -- WHICH qubits sit next to which -- that message passing preserves. The GNN's large jump specifically on floor-collapse cases, rather than a uniform improvement everywhere, is consistent with it picking up on a structural signature in the route graph that no single scalar feature could encode.

Entry 041 — Conclusion

The GNN was worth building now, at 1,376 circuits, rather than waiting for the low thousands originally planned: it already beats both the closed form and the tabular baseline by a wide margin, especially on exactly the failure mode (floor-collapse) neither could explain. That answers the diagnostic question this entry was built to answer -- the project was NOT dataset-starved for a first useful GNN. The floor-collapse gap the closed-form model treated as a hard ceiling (Entry 036) is not actually a hard ceiling for a model that sees the real route graph.

4. Honest Caveats

This is a first prototype: 16-dim hidden states, 2 message-passing rounds, 180 epochs, no hyperparameter search, and v4.1's own prediction is still an input feature rather than the GNN reasoning entirely from scratch. Five-fold CV on 1,376 examples with 70 floor-collapse cases (roughly 14 per fold) is a real but still modest sample for the exact subset that matters most -- the 6.22-point floor-collapse MAE should be read as promising, not final. The natural next check is whether it holds up as the dataset continues to grow.

5. Next Step

Grow the dataset further with the GNN now in place as the metric to track (not just the tabular baseline), watching specifically whether floor-collapse MAE keeps falling -- and consider a small ablation removing v4.1's prediction as an input feature, to see how much of the GNN's edge is genuinely new structural learning versus a residual correction on top of the closed form.

Research Log — Index Addendum (Entry 041)

Entry	Title	Date	Status
Entry 041	The First GNN (JAX/Optax, message passing over real routes)	Aug 16, 2026	Complete

Scripts: entry041_build_graphs.py, entry041_train_gnn.py. Data: quantumbridge_data/entry041_graph_dataset.json (1,376 graphs), quantumbridge_data/entry041_gnn_folds.json, quantumbridge_data/entry041_gnn_results.json. GNN beats v4.1 (63% lower floor-collapse MAE) and the tabular baseline (33% lower overall MAE) -- the biggest single-entry improvement since Entry 032.